

Casing Design Reference Manual

Document Information

Field	Value
Document Number	CASING-DES-2024-001
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Classification	Drilling Engineering Reference
Applicable Basins	Permian (Midland/Delaware), Eagle Ford, Williston (Bakken)

1. Casing String Types and Specifications

1.1 Conductor Casing — 20” OD

Parameter	Specification
Outer Diameter	20.000 in
Nominal Weight	94 lb/ft
Grade	H-40 or J-55
Connection	Buttress Thread (BTC)
Setting Depth	80–150 ft (driven or drilled/cemented)
Primary Function	Structural support, divert shallow gas, prevent washout

Conductor casing is the first string set. In the Permian Basin, conductor is typically driven to refusal in surface sand/shale formations. In the Williston Basin (Bakken wells), glacial till may require drilling and cementing the conductor to 100–150 ft.

1.2 Surface Casing — 13-3/8” OD

Parameter	Specification
Outer Diameter	13.375 in
Nominal Weight	54.5–68 lb/ft
Grade	J-55 or L-80
Connection	Buttress Thread (BTC) or ST&C

Parameter	Specification
Setting Depth	2,000–3,100 ft (varies by basin)
Primary Function	Protect freshwater aquifers, BOP support, kick tolerance

Surface casing setting depths by well (from `formation_tops.csv`):

Well	Formation at Shoe	Setting Depth (MD)	Setting Depth (TVD)
PERM-2847	Surface Casing / Spraberry top	2,500 ft	2,500 ft
PERM-3201	Surface Casing / Bone Spring Upper top	3,000 ft	3,000 ft
EF-1105	Surface Casing / Pearsall Shale top	2,000 ft	2,000 ft
BAK-0892	Surface Casing / Bakken top	2,500 ft	2,500 ft
PERM-4102	Surface Casing / Spraberry top	2,600 ft	2,600 ft
PERM-4305	Surface Casing / Bone Spring Upper top	2,800 ft	2,800 ft
EF-2201	Surface Casing / Pearsall Shale top	2,100 ft	2,100 ft
BAK-1456	Surface Casing / Bakken Upper top	2,600 ft	2,600 ft
DL-0773	Surface Casing / Bone Spring Upper top	3,100 ft	3,100 ft

1.3 Intermediate Casing — 9-5/8” OD

Parameter	Specification
Outer Diameter	9.625 in
Nominal Weight	40–53.5 lb/ft
Grade	L-80 or P-110
Connection	Buttress Thread (BTC) or Premium (VAM TOP)
Setting Depth	7,000–12,500 ft (formation dependent)
Primary Function	Isolate problem zones, manage pore pressure transitions

Intermediate casing is set across the transition from normal to abnormal pore pressure. Setting depth is determined by the fracture gradient of the weakest exposed formation versus the mud weight required to control the next open-hole section.

Intermediate casing setting depth criteria by formation:

Formation	Typical Pore Pressure (ppg)	Fracture Gradient (ppg)	Recommended Shoe Depth
Wolfcamp A	10.0–10.4	13.0–13.2	7,200–8,600 ft MD
Wolfcamp B	10.2–10.7	12.8–13.0	8,100–9,600 ft MD
Eagle Ford Shale	9.3–9.8	13.0–13.1	6,500–9,900 ft MD
Bakken	10.5–11.1	13.5–13.6	9,200–9,600 ft MD
Bone Spring Upper	9.9–10.6	13.4–13.6	8,200–8,800 ft MD
Bone Spring Lower	10.4–12.7	13.0–14.1	8,500–12,200 ft MD
Spraberry	9.2–9.6	13.5–13.6	5,500–5,600 ft MD
Dean	9.5–9.9	13.2–13.3	7,000–7,300 ft MD
Austin Chalk	9.5–10.0	12.8–12.9	9,800–13,000 ft MD
Three Forks	11.0–11.6	13.0–13.1	9,500–10,400 ft MD

1.4 Production Casing — 5-1/2” OD

Parameter	Specification
Outer Diameter	5.500 in
Nominal Weight	17–23 lb/ft
Grade	P-110 or Q-125
Connection	Premium (VAM TOP, TenarisHydril Blue, Hunting SEAL-LOCK)
Setting Depth	TD (total depth, through lateral in horizontal wells)
Primary Function	Wellbore integrity for production, completion isolation

Production casing must withstand burst pressures during stimulation (frac operations up to 10,000 psi treating pressure) and collapse pressures during production drawdown. Premium connections are mandatory for horizontal wells to ensure gas-tight seals through the build section.

2. Casing Grade Specifications

2.1 J-55

Property	Value
Minimum Yield Strength	55,000 psi
Minimum Tensile Strength	75,000 psi
Collapse Resistance (5-1/2", 17 lb/ft)	4,740 psi
Burst Resistance (5-1/2", 17 lb/ft)	6,190 psi
Body Yield (5-1/2", 17 lb/ft)	367,000 lb
Typical Application	Conductor, surface casing in low-pressure environments
Heat Treatment	Normalized or normalized and tempered
H ₂ S Service	Not recommended (non-sour service only)

2.2 L-80

Property	Value
Minimum Yield Strength	80,000 psi
Maximum Yield Strength	95,000 psi
Minimum Tensile Strength	95,000 psi
Collapse Resistance (9-5/8", 47 lb/ft)	5,310 psi
Burst Resistance (9-5/8", 47 lb/ft)	6,870 psi
Body Yield (9-5/8", 47 lb/ft)	1,086,000 lb
Typical Application	Surface and intermediate casing, sour service environments
Heat Treatment	Quenched and tempered
H ₂ S Service	Suitable per NACE MR0175 / ISO 15156

2.3 P-110

Property	Value
Minimum Yield Strength	110,000 psi
Maximum Yield Strength	140,000 psi
Minimum Tensile Strength	125,000 psi
Collapse Resistance (5-1/2", 23 lb/ft)	10,430 psi
Burst Resistance (5-1/2", 23 lb/ft)	12,960 psi
Body Yield (5-1/2", 23 lb/ft)	694,000 lb
Typical Application	

Property	Value
Heat Treatment	Intermediate and production casing, high-pressure wells Quenched and tempered
H ₂ S Service	Not recommended without special processing

3. Cement Program

3.1 Lead Slurry Design

Parameter	Specification
Slurry Weight	12.5–13.5 ppg
Yield	1.50–1.75 ft ³ /sack
Thickening Time	4–6 hours (BHCT dependent)
Compressive Strength (24 hr)	≥ 500 psi
Free Water	≤ 0.5 mL
Fluid Loss	≤ 200 mL/30 min
Additives	Extenders (bentonite, pozzolan), retarders, fluid loss agents

3.2 Tail Slurry Design

Parameter	Specification
Slurry Weight	15.8–16.4 ppg
Yield	1.15–1.22 ft ³ /sack
Thickening Time	3–5 hours (BHCT dependent)
Compressive Strength (24 hr)	≥ 2,000 psi
Free Water	0 mL
Fluid Loss	≤ 50 mL/30 min
Additives	Dispersants, retarders, gas migration agents, fluid loss agents

3.3 Top of Cement (TOC) Requirements

Casing String	TOC Requirement	Regulatory Basis
Conductor (20")	Surface	State regulations
Surface (13-3/8")	Surface (full returns)	40 CFR 146 / state UIC rules
Intermediate (9-5/8")	500 ft above previous shoe	API RP 10B-2, operator policy

Casing String	TOC Requirement	Regulatory Basis
Production (5-1/2")	Above kickoff point (KOP)	Operator policy, frac isolation

3.4 Pressure Test Criteria

Test	Pressure	Hold Time	Pass Criteria
Surface casing shoe test	FIT to 14.0 ppg EMW	10 minutes	$\leq 5\%$ pressure decline
Intermediate casing shoe test	FIT to next section MW + 0.5 ppg	10 minutes	$\leq 5\%$ pressure decline
Production casing pressure test	80% of burst rating or max treating pressure	30 minutes	$\leq 5\%$ pressure decline
Casing lap test (liner)	500 psi above max expected annular pressure	10 minutes	$\leq 5\%$ pressure decline

4. Casing Wear Limits and Inspection Criteria

4.1 Wear Classification

Wear Category	Wall Loss (%)	Action Required
Class A (New)	0%	No restrictions
Class B (Good)	$\leq 20\%$	Full service, document wear
Class C (Fair)	20–30%	Reduced pressure rating, monitor
Class D (Marginal)	30–40%	Restricted service, engineering review
Reject	$> 40\%$	Remove from service

4.2 Inspection Methods

Method	Application	Detection Capability
Electromagnetic (EM)	Internal/external wall loss	$\pm 5\%$ wall thickness
Ultrasonic (UT)	Precise wall thickness	$\pm 0.5\%$ wall thickness
Caliper (multi-finger)	Internal diameter profiling	± 0.001 in
Visual / borescope	Surface defects, corrosion	Qualitative

4.3 Wear Prediction

Casing wear rate depends on rotating hours, dogleg severity, and tool joint hardfacing:

- Wear rate (in/hr) $\approx 0.0001 \times \text{DLS } (^\circ/100\text{ft}) \times \text{RPM} / 100$
 - Maximum allowable wear before re-evaluation: 25% of nominal wall thickness
 - Wear monitoring interval: every 500 rotating hours or at each casing point
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5. Casing Running Procedures

5.1 Pre-Run Checklist

1. Verify casing tally matches well plan (joint count, weights, grades)
2. Drift all joints with API drift mandrel (5-1/2": 4.653 in drift diameter)
3. Inspect threads for damage — reject any joint with thread galling or cross-threading
4. Verify float equipment (float collar, float shoe, centralizers) quantity and placement
5. Confirm centralizer placement plan: one per joint through build section, every third joint in vertical
6. Verify fill-up schedule: fill every 5 joints to prevent collapse from hydrostatic imbalance

5.2 Makeup Torque Specifications

Casing Size	Connection	Optimal Torque (ft-lb)	Min Torque (ft-lb)	Max Torque (ft-lb)
20" (94 lb/ft)	BTC	18,500	15,700	21,300
13-3/8" (68 lb/ft)	BTC	13,200	11,200	15,200
9-5/8" (47 lb/ft)	BTC	8,800	7,500	10,100
9-5/8" (47 lb/ft)	VAM TOP	12,500	10,600	14,400
5-1/2" (23 lb/ft)	BTC	5,400	4,600	6,200
5-1/2" (23 lb/ft)	VAM TOP	7,800	6,600	9,000
5-1/2" (17 lb/ft)	BTC	4,100	3,500	4,700

5.3 Running Speed Limits

Condition	Maximum Running Speed
Open hole, no restrictions	3 joints/min (≈ 120 ft/min)
Through tight spots or doglegs	1 joint/min (≈ 40 ft/min)
Through BOP stack	Controlled, no spinning
Last 500 ft to TD	1 joint/min, circulate bottoms-up

5.4 Post-Run Verification

1. Circulate bottoms-up (minimum $1.5\times$ annular volume)
2. Reciprocate casing ± 30 ft during circulation (if hole conditions allow)
3. Pressure test casing to specified test pressure before cementing
4. Verify casing is at planned depth (± 5 ft tolerance)
5. Record final hanging weight and compare to calculated string weight ($\pm 10\%$ tolerance)

6. Formation-Specific Casing Design Considerations

6.1 Wolfcamp A/B (Permian Basin)

- Pore pressure: 10.0–10.7 ppg; fracture gradient: 12.8–13.2 ppg
- Intermediate casing (9-5/8") set at Wolfcamp A top (7,200–8,600 ft MD)
- Production casing (5-1/2") P-110 required for frac operations (10,000+ psi treating)
- H₂S potential: low to moderate — L-80 acceptable for intermediate string

6.2 Eagle Ford Shale

- Pore pressure: 9.3–9.8 ppg; fracture gradient: 13.0–13.1 ppg
- Intermediate casing set at Eagle Ford top (6,500–6,600 ft MD)
- Austin Chalk above Eagle Ford may require lost circulation material (LCM)
- Production casing: P-110 with premium connections for horizontal lateral

6.3 Bakken / Three Forks (Williston Basin)

- Pore pressure: 10.5–11.6 ppg; fracture gradient: 13.0–13.6 ppg
- Higher pore pressures require L-80 minimum for intermediate casing
- Three Forks dolomite is abrasive — monitor casing wear during drilling
- Long horizontal laterals (10,000+ ft) require torque-and-drag analysis for casing running

6.4 Bone Spring (Delaware Basin)

- Pore pressure: 9.9–12.7 ppg (wide range, depth dependent)

- Lower Bone Spring may approach fracture gradient — careful mud weight management
- Intermediate casing set at Bone Spring Lower top or within Bone Spring interval
- DL-0773 example: Bone Spring Lower pore pressure 12.1 ppg vs fracture gradient 14.1 ppg

6.5 Spraberry / Dean (Midland Basin)

- Pore pressure: 9.2–9.9 ppg; fracture gradient: 13.2–13.6 ppg
- Relatively benign pressure environment — J-55 or L-80 surface casing acceptable
- Dean sandstone may produce water — cement integrity critical at Dean shoe
- Spraberry siltstone can cause differential sticking — maintain adequate overbalance

Appendix A: API Casing Dimensions Quick Reference

OD (in)	Weight (lb/ft)	ID (in)	Drift (in)	Wall (in)
20.000	94.00	19.124	18.936	0.438
13.375	54.50	12.615	12.459	0.380
13.375	68.00	12.415	12.259	0.480
9.625	40.00	8.835	8.679	0.395
9.625	47.00	8.681	8.525	0.472
9.625	53.50	8.535	8.379	0.545
5.500	17.00	4.892	4.653	0.304
5.500	20.00	4.778	4.539	0.361
5.500	23.00	4.670	4.431	0.415

Reference Standards: API 5CT (Casing and Tubing), API RP 10B-2 (Cement Testing), API RP 65-2 (Cementing Annular Spaces), NACE MR0175/ISO 15156 (Sour Service Materials)